

## Japan Quant Strategy

Beta x Momentum reversal – Backlash against overheating and tactical appeal of bank stocks during the Momentum recovery phase

*This week's sharp decline in the Beta x Momentum index is a technical-led adjustment in reaction to overheating from April. Following a  $-5\sigma$ -level crash, there is a high probability that the Momentum effect will remain subdued for about a month. As such, a gradual shift toward Value stocks appears to be an appropriate strategy for the time being. Among these, the banking industry, particularly regional banks, which combine both Momentum and Value, could serve as an effective tactical hedge against the unwinding risks in AI/semiconductor-related stocks.*

### 1-Beta/Momentum's reverse rotation appears temporary

This week, the Nikkei Average briefly dipped below the 60,000 mark, as a broad range of AI/semiconductor-related stocks, which had been driving the market overall, were down significantly. Meanwhile, as the US semiconductor stock index (SOX) rebounded, dip-buying emerged in Japan as well, and today (May 21) the Nikkei Average regained the 61,500 level. Among market participants, while the long-term bullish story of AI-related stocks, which have been rising at a rapid pace, remains largely intact, a cautious view that price levels have become overextended in the short term and that a correction is likely to continue for the time being is gaining traction.

From a style factor perspective, this sharp shift can be interpreted as a slowdown in the Beta  $\times$  Momentum strategy, which had significantly outperformed the market during the relief rally in April. We have constructed a 3x3 portfolio that double-sorts the TOPIX 500 universe with Beta and Momentum (12 months, excluding the most recent month). Looking at the excess return relative to TPX for each bucket, we can confirm that there is an irregular downward refraction in the cumulative excess return for the High Beta  $\times$  High Momentum portfolio. Meanwhile, the Low Beta  $\times$  Low Momentum group has begun to rebound and the market as a whole has been on a reversal trend since the end of last week.

#### Global Quantitative and Derivatives Strategy

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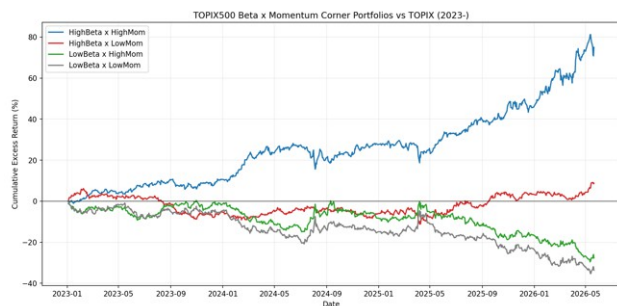
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**Figure 1: TOPIX 500-based Beta x Momentum portfolio excess returns (segmented into 3x3 groups based on Beta and 12-month returns; only the four main series are shown in figure) – High Beta & High Momentum fell sharply**



Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

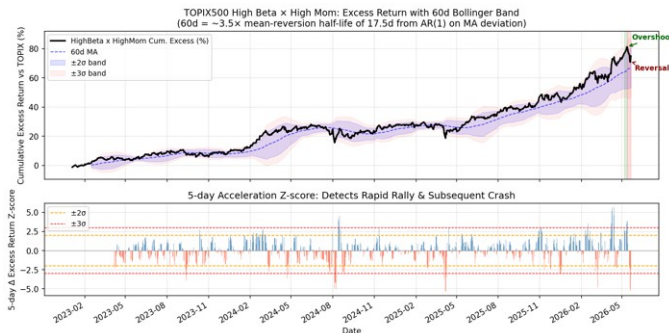
This reversal is characterized by the fact that the correction occurred rapidly within a short period of time. When the excess return (weekly) of High Beta x High Momentum strategies are converted to z-scores, it surges to  $+3.9\sigma$  between May 7 and 13, then reverses course to around  $-5.3\sigma$  between May 14 and 20. If we take this at face value, this suggests that the market phase that has been controlled by High Beta x High Momentum has reached an inflection point. However, what we should not overlook here is that the sharp rise itself since early May has, statistically speaking, already reached a point where it has run its course.

Specifically, we calculated price bands for the cumulative excess return series of the portfolio in question using a window based on the mean reversion half-life (17.5 days, based on an autoregressive model), specifically, 3.5 half-lives, which is the period required for approximately 95% of the deviation to disappear (i.e.,  $17.5 \times 3.5 \approx 60$  days). What we found was that the cumulative excess return had consistently remained above the  $\pm 2\sigma$  band since mid-April. Statistically speaking, it is more reasonable to interpret the recent pullback as a reaction to market overheating rather than as a sign of a fundamental shift. In other words, this  $-5\sigma$ -level shock can be viewed as the result of a reversal of the significant outperformance seen since April.

In addition, a global rise in long-term interest rates, including in Japan, has been cited as a factor contributing to the slowdown in the Beta x Momentum strategy. Although it is undeniable that the rise in long-term interest rates is not a positive factor for equity risk premiums, the mechanism behind this is primarily multiple compression driven by the discount rate effect, a characteristic that tends to manifest most strongly in growth and quality stocks, where relatively high valuations have become the norm. It would be an overly simplistic assessment to attribute the market-wide risk-off sentiment entirely to the rise in long-term interest rates.

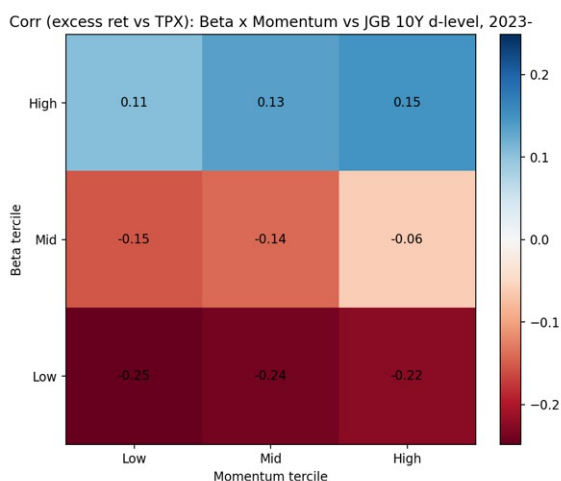
To support this point, we estimated the correlation between the excess return of the Beta x Momentum 3x3 portfolio, 10-year JGB yields, and the JGB 2s10s curve (focusing on periods of inflation and rising Japanese stock prices from 2023 onward). Examining the correlation with the rise in 10-year JGB yields by Beta type, the average for High-Beta banks is  $+0.13$ , for Mid-Beta banks is  $-0.12$ , and for Low-Beta banks is  $-0.24$ , indicating a monotonic relationship with respect to the Beta direction. On the other hand, the correlation difference in the Momentum direction within the same Beta line was only around 0.02 to 0.09, suggesting that its impact is secondary. On average, periods of rising long-term interest rates have tended to favor High-Beta stocks. However, there is no evidence to suggest that rising interest rates directly cause a reversal in Beta.

**Figure 2: The Outperformance of High Beta and High Momentum – In early May, the Momentum effect rapidly added to the Beta effect, and it appears that about half of the recent overheating has now subsided**



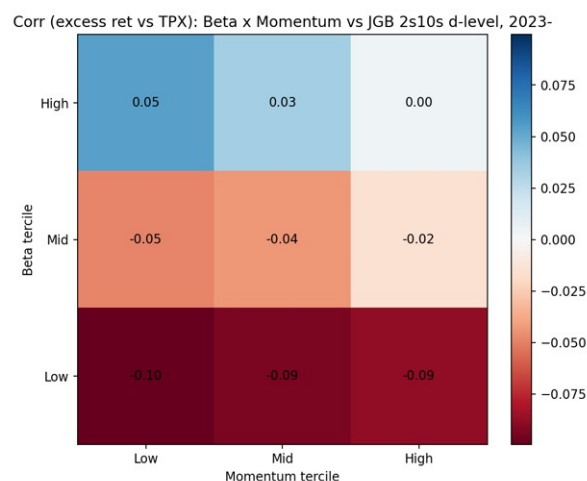
Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

**Figure 3: TOPIX 500-based Beta x Momentum (3x3 split) portfolio excess return and 10-year JGB yield correlation (since 2023, as of May 20) – During the stock market rally in 2023, the outperformance of High-Beta stocks and the rise in the 10-year JGB yield coexisted**



Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

**Figure 4: TOPIX 500-based Beta x Momentum (3x3 split) portfolio excess return and JGB 2s10s curve correlation (from 2023, as of May 20) – A steeper curve acts as a tailwind for Beta and is unlikely to be the primary driver of the recent Beta unwinding**



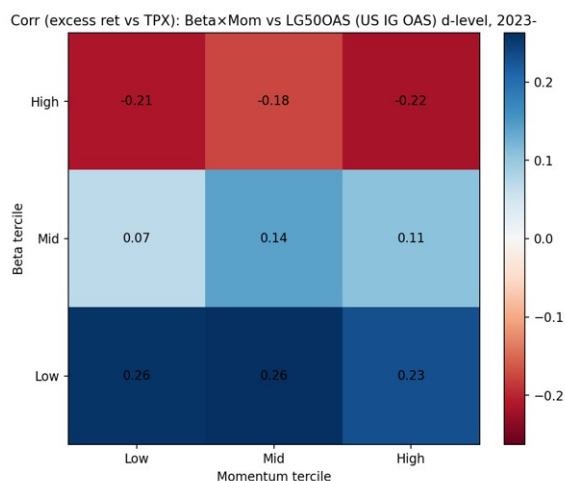
Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

If this correlation pattern is extrapolated, in order for a full-fledged reversal from High Beta to Low Beta to take hold, a shift in the external environment significant enough to trigger a sell-off in Beta itself is required. Specifically, it would require the emergence of stress in global credit markets or a sharp appreciation of the yen against the dollar. However, there are no signs of that at this point. The global credit market appears stable, with no signs of systemic stress. As for the forex market, the USD/JPY rate dipped to the ¥156 level immediately after the intervention by the Japanese government and Bank of Japan (BoJ) but has since recovered to the ¥159 level. Speculative investors, including CTAs, have refrained from unwinding their long positions in USD/JPY and have actually resumed building up their long positions. It would be difficult to say that any of them are clearly leaning toward pessimism.

Based on the above, it appears that the recent sharp rise in long-term interest rates has forced adjustments primarily on the Momentum side rather than the Beta side. By definition, Beta reflects the overall market direction (i.e., macro fundamentals). In that respect, no significant shifts have been observed at this time. In contrast, Momentum reflects the accumulation of funds that have until now been concentrated on popular stocks. We can thus conclude that the Momentum effect, which had been riding on top of the Beta effect from April through May, has expanded due to leverage. The sudden rise in long-term interest rates has triggered a sharp unwinding of positions in Momentum stocks with stretched valuations and crowded positions, and High-Beta stocks were also caught up in the sell-off. In the short term, the collapse of excess returns for High Beta is probably the result of this supply-demand chain reaction.

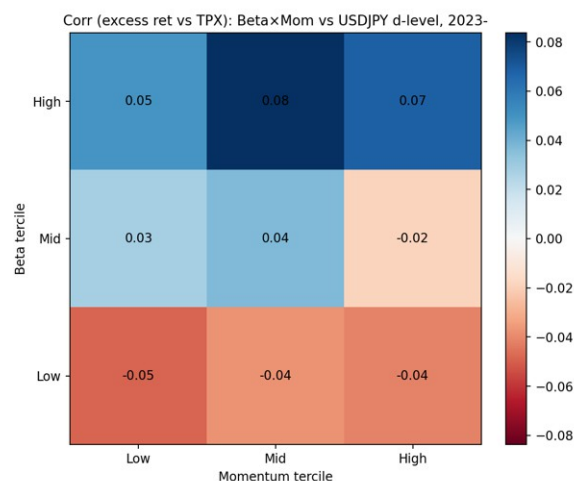
Therefore, it makes more sense to interpret this occurrence not as a correction driven by a shift in fundamentals but rather as a self-driven correction driven by technical factors. In other words, it is a sharp price movement resulting from a concentration of position unwinding over a short period. Rather than hastily concluding that the reversal in Beta x Momentum signals a major shift in the market, we see this as a temporary phenomenon resulting from a sudden reaction to the market overheating.

**Figure 5: TOPIX 500-based Beta x Momentum (3x3 split) portfolio excess return and US credit spreads correlation (since 2023, as of May 20) – Beta positions are vulnerable to a global deterioration in credit conditions**



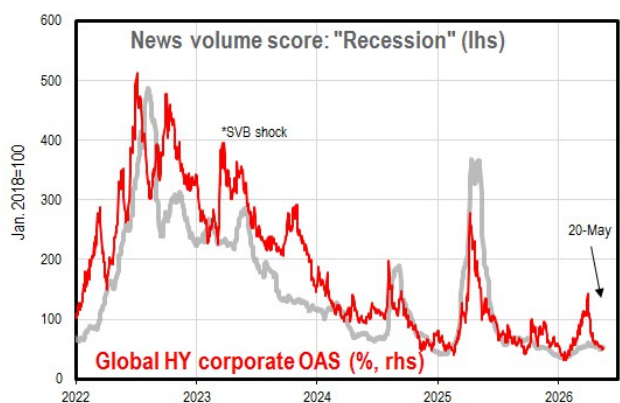
Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

**Figure 6: TOPIX 500-based Beta x Momentum (3x3 split) portfolio excess return and USD/JPY correlation (since 2023, as of May 20) – While the correlation is nearly zero, excessive yen appreciation tends to trigger Beta selling**



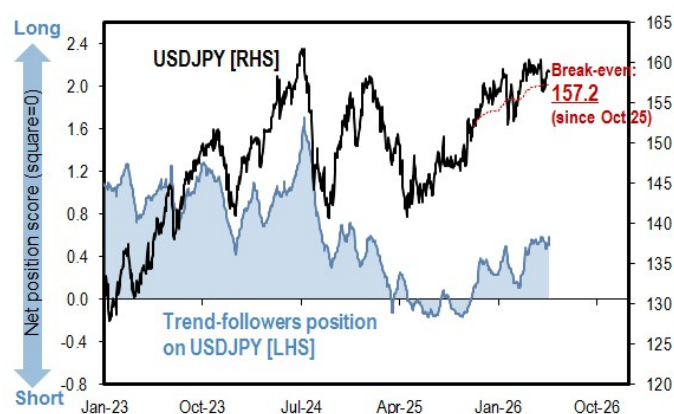
Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

**Figure 7: Number of media reports score (global recession) and global high-yield spread (OAS)**



Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

**Figure 8: CTA's USD/JPY positions – Limited decline in USD/JPY after intervention, and CTAs did not face a forced unwinding**

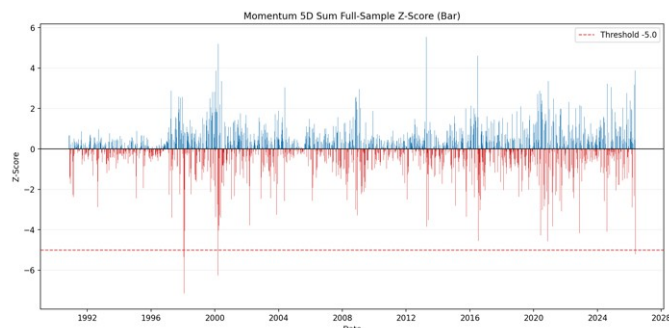


Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

## 2-Prescription for the post-crash healing phase

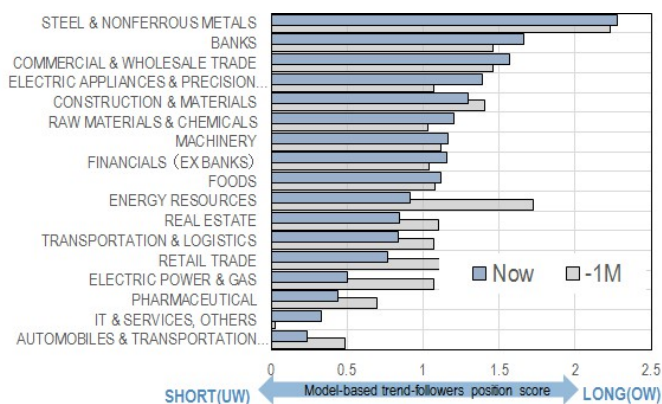
Circling back to the Momentum factor itself, which is the trigger for the adjustment we have been discussing so far, as of May 20 the weekly change score for the Momentum factor stood at  $-5.2\sigma$ . By definition, we cannot deny that an event classified as a “Momentum crash” has occurred. Reviewing past instances where similar shocks occurred reveals a recurring pattern in which the Momentum factor (Long/Short) has struggled to recover for roughly one month following the shock. We need to be clearly aware of this point as a matter of caution this time around as well.

**Figure 9: Japan Momentum factor (1) – A “crash” exceeding  $-5\sigma$  occurred in the cumulative 5-day return (as of May 20)**



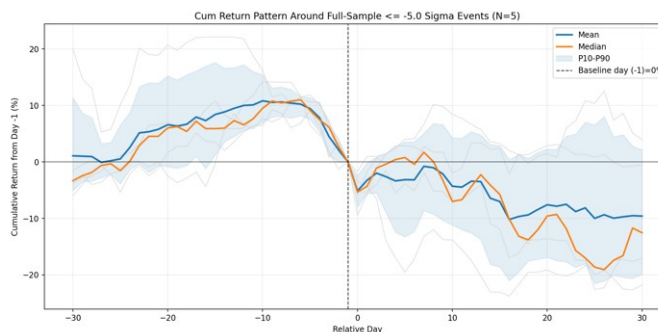
Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

**Figure 11: Cash equity Trend-Followers position by sector (aggregated by TPX17 sector, for covering TPX500 constituent stocks): As of May 20 – Steel, non-ferrous metals, and banking sectors are leading the way**



Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

**Figure 10: Japan Momentum factor (2) – Momentum factor performance after a “crash” exceeding  $-5\sigma$**



Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

**Figure 12: Example of the Momentum effect (Long/Short) by sector – Since the second half of last year, the steel and non-ferrous metals sectors have been driven primarily by a small number of rising stocks, while the banking sector may be showing broad-based resilience across a wide range of stocks**



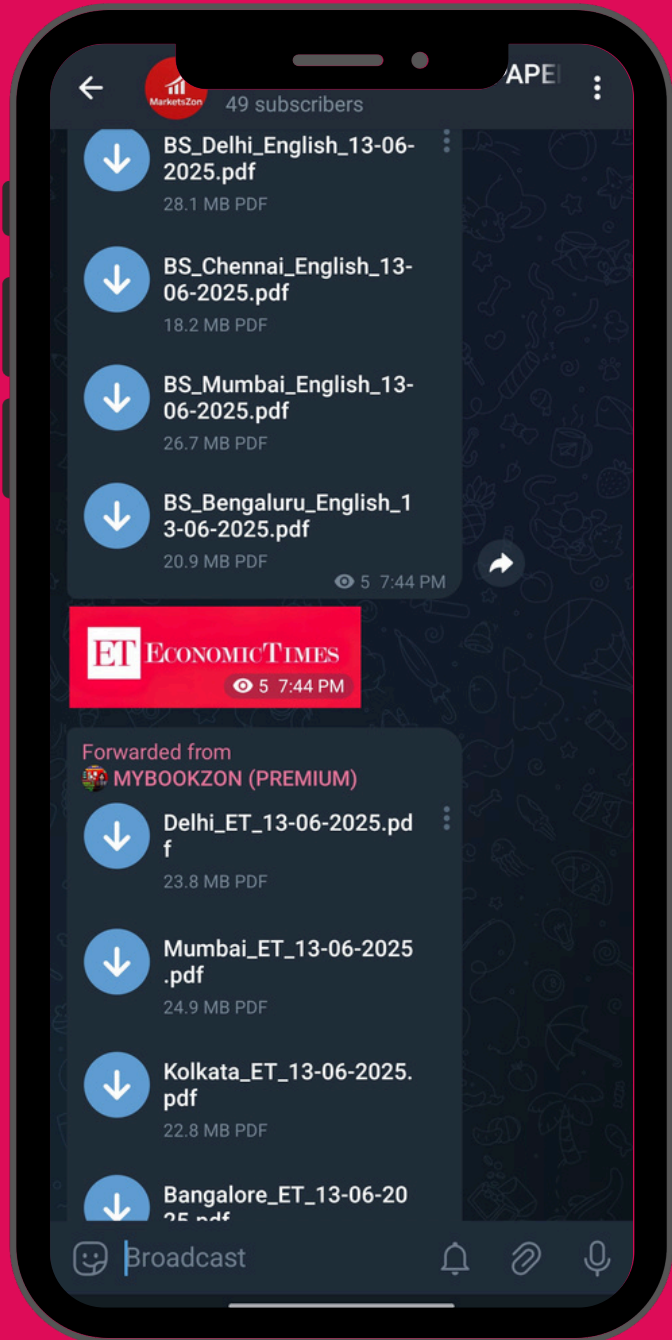
Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

Of course, when comparing this with past crashes, there are noticeable differences. In general, when Momentum crashes occur, they are often coupled with a complete reversal (a 180-degree shift) in the fundamental trends or investment themes that had previously driven the buying of popular stocks and selling of unpopular ones, or in the investment themes themselves. In contrast, the current crash differs in that there are no clear factors that would trigger a reversal in the fundamental economic environment itself. As mentioned above, whether we look at global economic indicators, earnings outlooks, credit conditions, or exchange rates, no changes have been observed that would instantly rewrite market assumptions. This market shock was the result of technical factors that autonomously caused overheating and cooling, and it was somewhat different from a fundamentals-based crash.

Even so, we cannot ignore the lingering effects of a crash once it has occurred. For the time being (or at least a month or so), it would be reasonable to factor the risk of a slowdown in the momentum effect into the base case. Based on this premise, a useful historical precedent is the Momentum crash that preceded the collapse of the IT bubble, when market conditions were somewhat similar to those of the current Japanese stock market. Looking at the performance of different sectors at that time, the Japanese market saw weakness in banking, IT services, and trading companies, and defensive stocks dominated the market.

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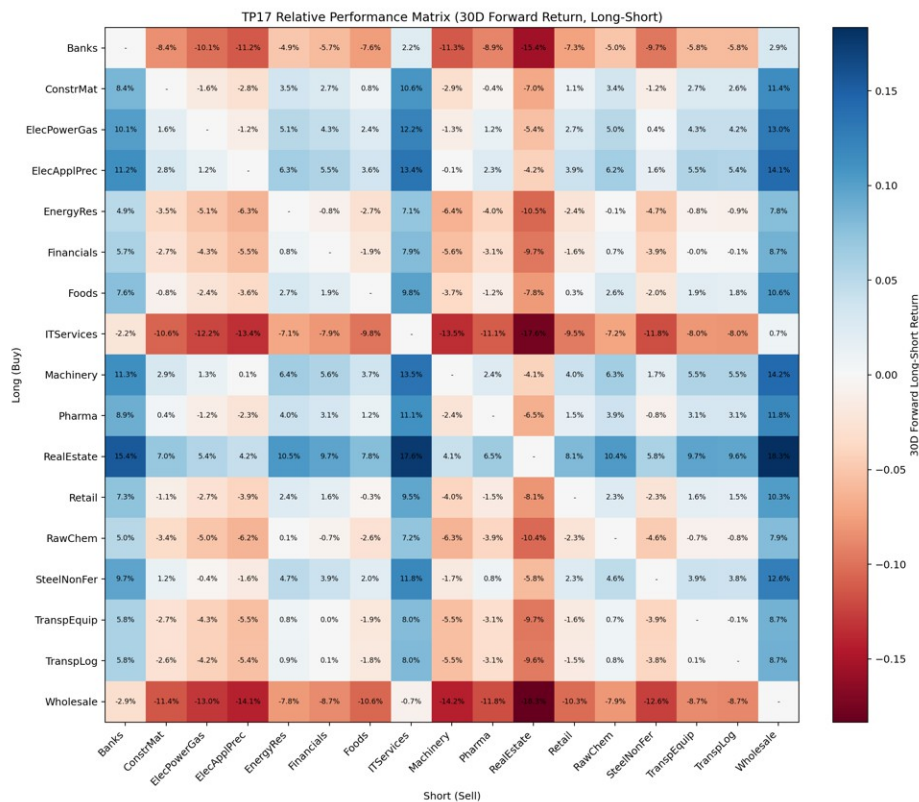


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In this regard, Japan's macroeconomic and interest rate environment today is vastly different from what it was back then. In particular, given that the BoJ is currently raising interest rates, there is a stark difference from the period when the economy was plagued by non-performing loans and a financial slump. We need to be cautious when applying past performance data by sector directly to current conditions. A more relevant point to consider is the performance of stock markets in developed countries at the time. Although growth sectors such as IT and telecommunications services saw sharp declines, defensive, value, and quality sectors, including consumer staples, banking, and pharmaceuticals, relatively outperformed.

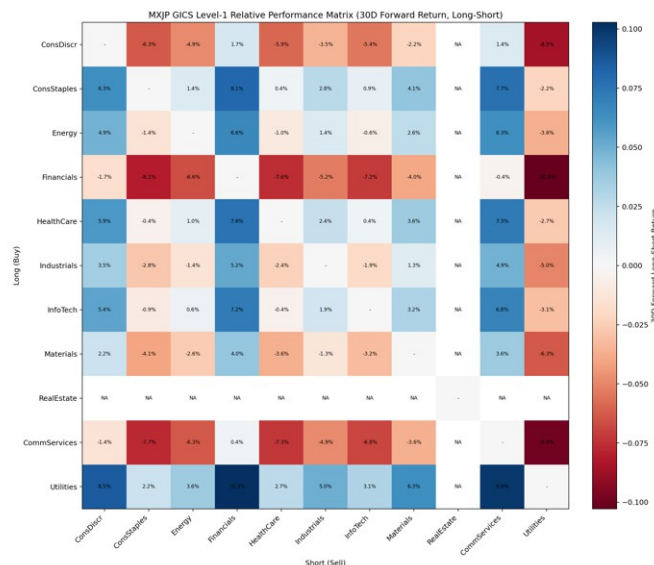
Applying this rule of thumb to the current situation, while the Momentum factor may still harbor the potential for volatility, the macro environment itself is not leaning toward pessimism. Therefore, rather than adopting an approach that involves completely overhauling portfolios by replacing the winners with losers all at once, selectively picking Momentum stocks that have not become overly crowded from among value and cyclical sectors appears to be the most appropriate strategy under current conditions. In other words, the basic approach to weathering the post-crash recovery phase is to rotate a portfolio's style allocation toward a slight Value bias without completely abandoning Momentum.

**Figure 13: Relative sector performance of Japanese stocks following the Momentum crash on the eve of the collapse of the IT bubble in 2000 (One-month forward returns calculated using Long/Short sector strategies; based on the TPX 17 sectors) – Underperformance observed in the banking, IT services, and trading company sectors**



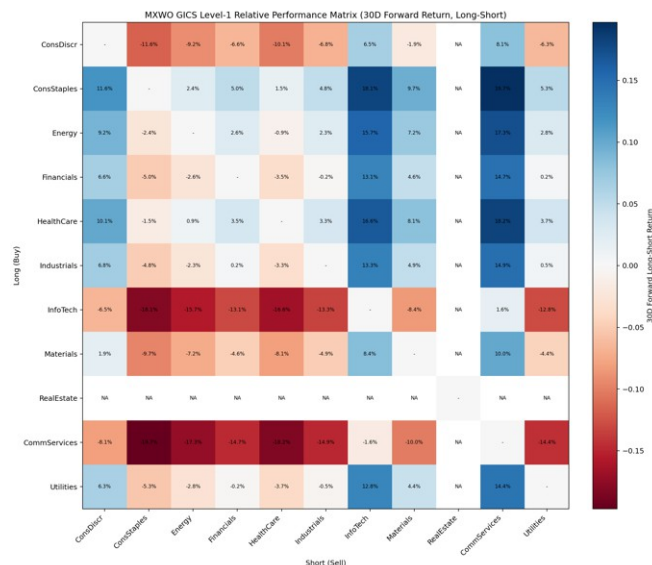
Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

Figure 14: Relative sector performance of Japanese stocks following the Momentum crash on the eve of the collapse of the IT bubble in 2000 (One-month forward returns calculated using Long/Short sector strategies; use of GICS-level 1 classification)



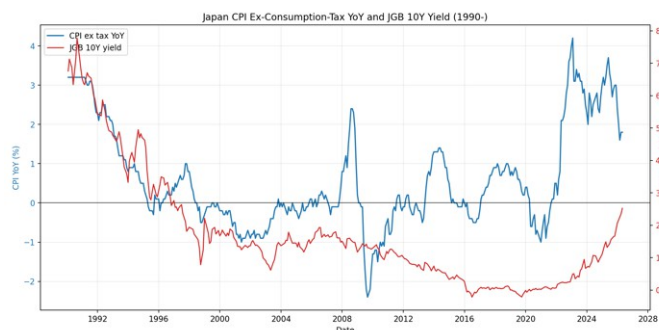
Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

Figure 15: Relative sector performance of developed market stocks following the Momentum crash on the eve of the collapse of the IT bubble in 2000 (One-month forward returns calculated using Long/Short sector strategies; use of GICS-level1 classification)



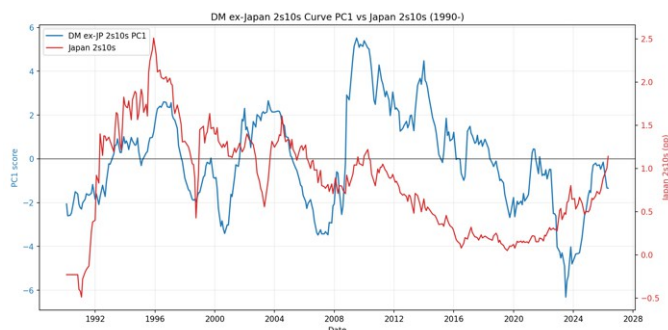
Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

Figure 16: Japan's CPI inflation rate and 10-year JGB yield – Since Japan was experiencing deflation and a zero interest rate policy (ZIRP) at the time of the collapse of the IT bubble in 2000, the current environment is different



Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

Figure 17: Common factors driving fluctuations in the 2-year/10-year yield curves of developed countries (PC1 component) and Japan's 2-year/10-year yield curve: Japan's yield curve did not steepen sufficiently at the time of the collapse of the IT bubble in 2000



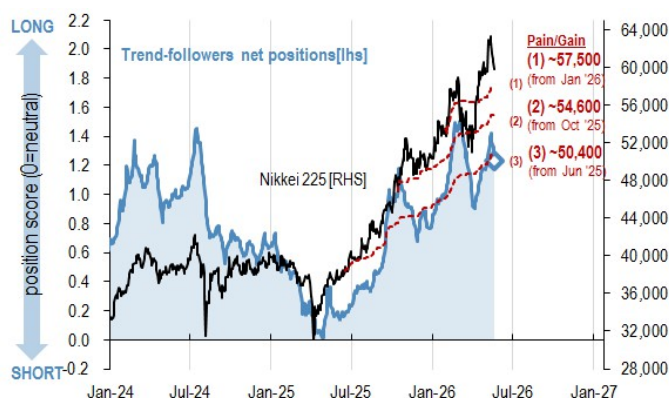
Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

### 3-The appeal of regional banks as Value winners

In this environment, we, the QDS team, are focusing on the banking sector, particularly regional banks, as an effective hedge for the time being. Faced with potential Momentum unwinding risks for AI/semiconductor-related stocks, banks can provide hedging protection through four distinct channels. We think it is highly likely that fund flow, interest rate scenarios, factor exposure positioning, and trends in earnings momentum will all act as a tailwind for bank stocks.

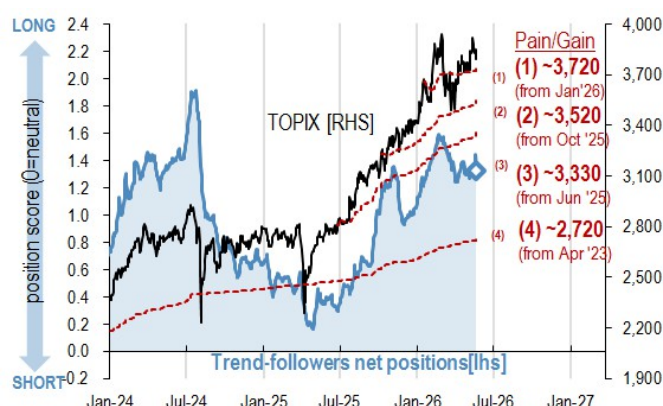
First, there are signs of a shift in the flow of funds among players in the global index sector. CTAs, which had built up excessive long positions in Nikkei Average futures, are now conducting a trial rotation into TOPIX futures, where position unwinding has lagged behind. Although this is still purely a technical reaction, there are signs of a partial rebalancing of regional allocations, heavily weighted toward AI/semiconductor-related sectors. This week, profit-taking in NKY futures, KOSPI 200 futures, and Taiwan Stock Index futures has accelerated, while long positions in TOPIX futures have increased. That we are currently in a period where fund flows can easily shift from high-growth stocks to large-cap Value stocks is likely to be a tailwind for traditional Value sectors, particularly the banking industry.

**Figure 18: CTA's Nikkei futures positions – Amid the turbulent market conditions marked by a sharp rebound in positions starting in April and the subsequent plunge in the index, CTAs have begun to show signs of “buyer fatigue”**



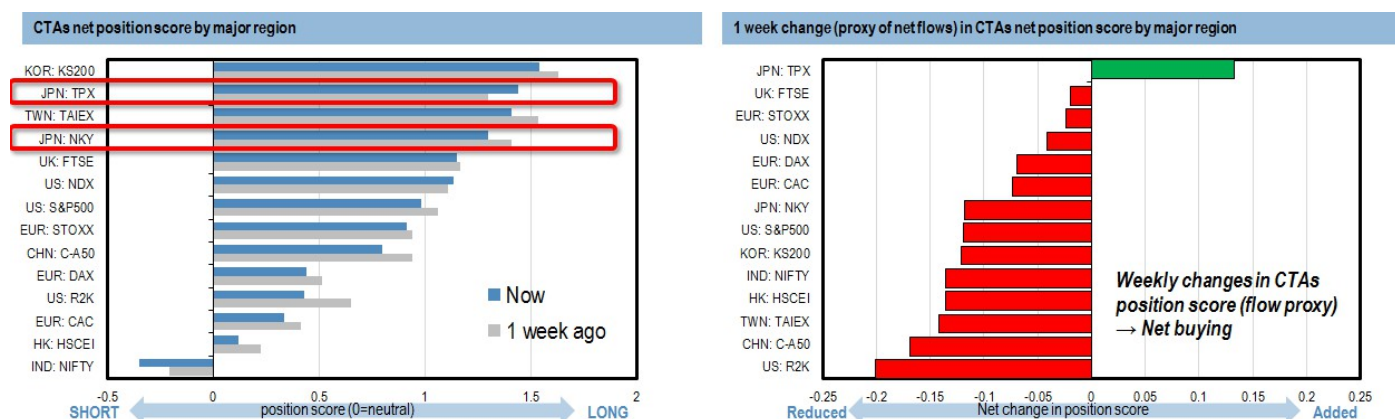
Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

**Figure 19: CTA's TPX futures positions – With the post-April rally driven by AI/semiconductor stocks, the TOPIX has lagged behind the Nikkei 225, and CTAs are finally showing signs of rebuilding their positions**



Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

**Figure 20: Comparison of CTA's net exposure to equity futures by region (using normalized scores, as of May 20) – NKY and TPX swapped places in the top tier of CTA preferences**

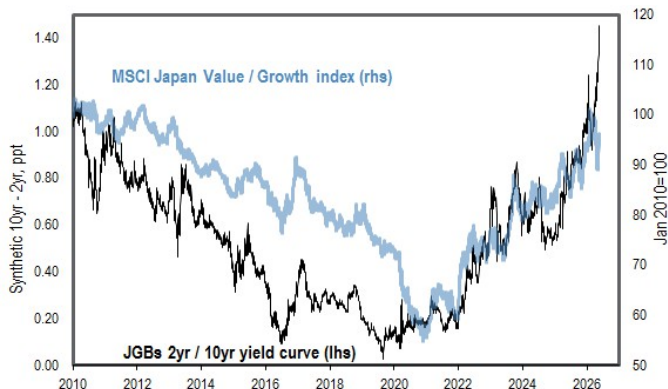


Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

Second, speculative positions in JGBs are already heavily pricing in the expectation that long-term interest rates will continue to rise. Both CTAs and macro hedge funds have been building bearish positions in JGBs, and the positioning of these top-down players is in line with the BoJ's hawkish policy stance and Japan's inflationary environment. Barring a complete reversal in the consensus regarding Japan's inflation and monetary policy, these positions will not easily collapse. Assuming a scenario in which the BoJ resumes raising interest rates from June, it is entirely plausible that bank stocks, which benefit from rising

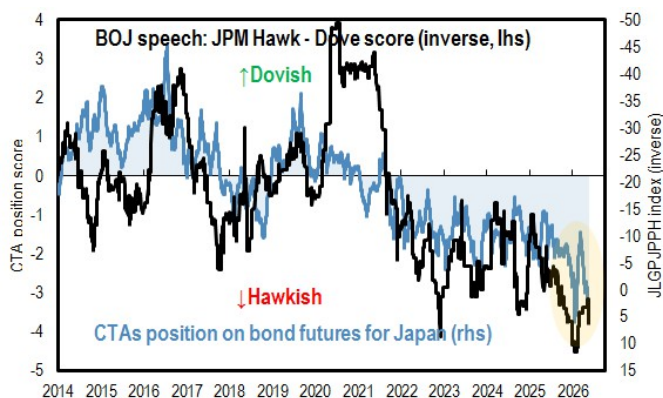
interest rates in terms of both earnings growth and share prices, will temporarily take the lead, allowing Value stocks to regain some ground against Growth stocks.

**Figure 21: 2yr/10yr JGB yield curve and MSCI Japan Value vs. Growth Index – Since April, a gap has emerged between the yield curve and stock price performance**



Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

**Figure 23: CTA's JGB-related positions and BoJ speech score – CTAs maintain largest-ever bearish JGB positions**



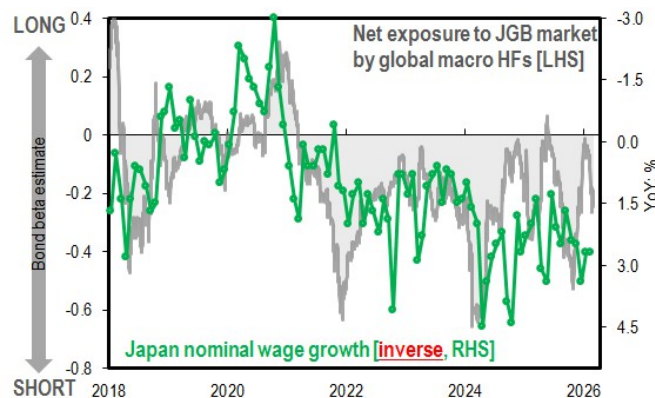
Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

**Figure 22: JPY 3m/10yr OIS curve (1y Z-score conversion) and Cash equity Trend-Followers' total banking position – Momentum player position generally matches the trend of interest rate changes**



Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

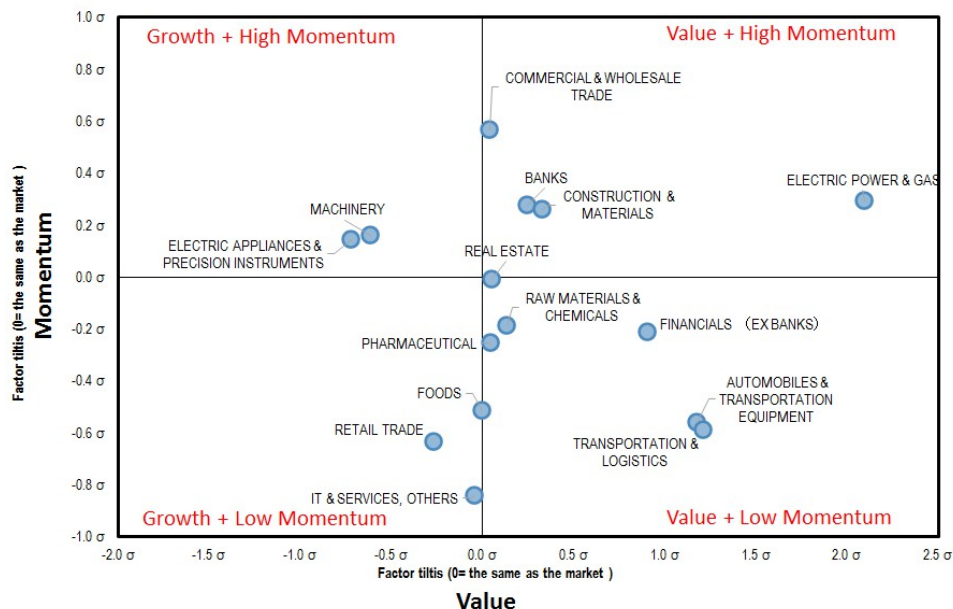
**Figure 24: Macro hedge funds' JGB-related positions and Japan's nominal wage growth rate – We estimate that they have once again begun building bearish positions on JGBs**



Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

Third, with regard to factor exposure, banks are in a favorable position from both an offensive and defensive standpoint. When estimating exposure to Value and Momentum by sector, the banking sector holds both moderate Momentum effects and moderate Value effects. Construction and materials are also cited as industries that show similar positioning. In the current market environment, our base case scenario is that Value stocks will outperform. However, to the extent that no major changes in fundamentals have been confirmed, it is difficult to determine in advance when growth, led by AI/semiconductor-related companies, will rebound. Given this situation, banks naturally meet the necessary criteria on a sector-by-sector basis, as they allow a portfolio to capture a gradual Momentum effect while at the same time avoiding an excessive tilt towards Value and making it vulnerable to a Growth rebound.

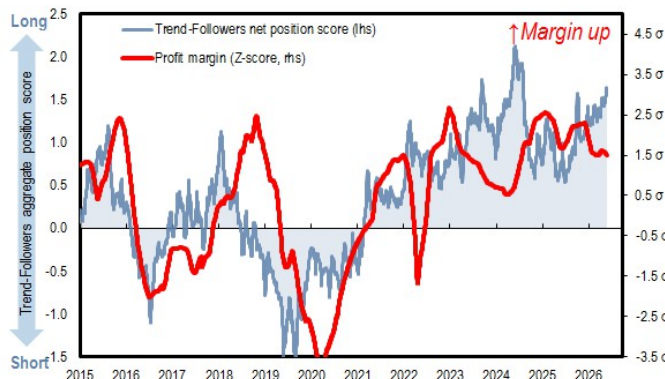
Figure 25: Value & Momentum factor loading scatter diagram by sector (based on TPX17, as of the end of April) – Banks hold a position with moderate exposure to both Momentum and Value (the same applies to Construction and Materials)



Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

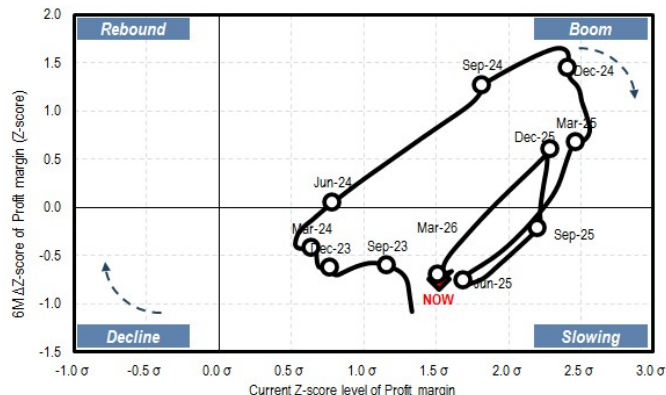
Fourth, trends in corporate earnings are also likely to act as a tailwind for bank stocks. Compared to the end of March, when the market had already factored in a deteriorating situation in the Middle East, as of May 21 the EPS revision index shows that the real estate and banking sectors are outperforming the precision equipment sector, centered on semiconductor manufacturing equipment. If margins in the banking sector begin to recover, the overall industry margin cycle could shift again from a slowdown to an expansion phase. A shift in the margin cycle often leads to upward revisions in bottom-up earnings forecasts, thereby providing momentum-seeking investors with a rationale for new purchases. Based on both fund flow and earnings, it is easy to see a pattern where positive factors for bank stocks are likely to accumulate.

Figure 26: Banking (TPX17 classification, aggregate value of applicable TPX500 constituents): Total positions of Cash equity Trend-Followers and NTM profit margin (converted to 2-year Z-score)



Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

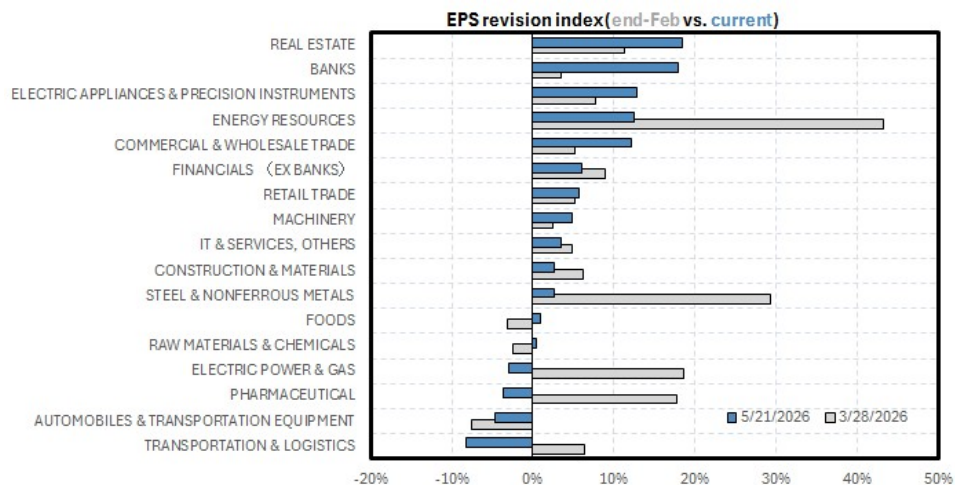
Figure 27: Profit margin cycle chart for the banking sector (TPX17 basis) (x-axis: Z-score of NTM profit margin, y-axis: 6-month difference) – Currently in the “Slowing” phase



Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

**Figure 28: EPS revision index by sector (comparison from end-March to May 21) – The geopolitical uncertainty of the Middle East conflict has been a major factor, but revisions for the banking sector, along with real estate and trading companies, have recently begun to improve**

Bloomberg consensus aggregate value is used



Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

Finally, we confirm our positioning of the banking sector at the individual stock level. For reference, when we select the top 50 stocks with the highest Momentum positions from the TOPIX 500 universe, we find that many are currently in what is known as a “pain trade” (i.e., unrealized losses). For stocks in a “pain-trade” situation, mechanical selling pressure resulting from position reduction is likely to remain high. In this context, the overall Momentum effect may continue to wane in the short term.

However, it is noteworthy that the top 50 stocks include many bank stocks, and that most of them have been able to avoid the “pain trade” situation. Most of the stocks that are currently avoiding “pain trades” are those that posted relatively modest returns during April’s rally compared to AI/semiconductor-related stocks. However, that has actually worked in their favor during this market crash, helping to minimize the damage. In addition, looking at the positions of individual stocks in the banking sector, the top Momentum positions include more regional banks than megabanks. In the event that some of the funds flowing out of AI/semiconductor-related sectors flow into the banking sector in search of Momentum, regional bank stocks, which have relatively small market capitalizations, are likely to be more susceptible to supply/demand dynamics, potentially leading to significant price fluctuations.

In general, we see a high probability of a period of lingering effects from the Momentum crash for the near term. We thus believe it is prudent to adjust the overall portfolio style to lean slightly toward Value while not completely abandoning Momentum. In addition, we believe that maintaining a tactical long bias toward the banking sector (particularly regional banks) will prove beneficial during the period until AI/semiconductor-related stocks regain some stability.

Figure 29: Cash equity Trend-Followers positions (Long-bias stocks, top 50 stocks): As of May 20

1. Universe is TOPIX 500 constituent stocks. 2. If the position shows a change in the opposite direction for two or more periods, including one-week, one-month, and three-month changes, then keep Momentum? If it occurs in three places, indicated by x respectively. 3. Shows unrealized gains (gain) or unrealized losses (pain) by comparing position's break-even point (three-month cumulative basis) with most recent share price.

| Name (LONG Top-1) |                           | Rank (within the universe) |                     | Trend Followers position score |          |           |                           |                         |                         |                         |                         |                         |                     | Price          |      | Pain/Gain Monitor(3M window) |                      |        |        | Abs. return % |      | Fundamentals |              |              |              |              |                    |  |
|-------------------|---------------------------|----------------------------|---------------------|--------------------------------|----------|-----------|---------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|---------------------|----------------|------|------------------------------|----------------------|--------|--------|---------------|------|--------------|--------------|--------------|--------------|--------------|--------------------|--|
| Code              | Company                   | Sector                     | -3M→-1M→<br>-1W→Now | Pos. Score<br>(+long, -short)  | (5Y Ytd) | Keep Mon? | Pos. Chg<br>(1Wk, 1M, 3M) | Momentum<br>/ Reversal? | Pos. Chg<br>/ Reversal? | Momentum<br>/ Reversal? | Pos. Chg<br>/ Reversal? | Momentum<br>/ Reversal? | Mkt Cap<br>(Bn JPY) | Price<br>(JPY) | Q/P? | Break-even<br>level (JPY)    | Distance to<br>B/E % | 1wk %  | 1M %   | 3M %          | Beta | 60D HVol     | P/E<br>(12m) | P/B<br>(12m) | ROE<br>(12m) | DY%<br>(12m) | Foreign<br>sales % |  |
| 8377              | HOKUOKU FINANCIAL (8377)  | Banks                      | 7→2→2→1             | 2.7                            | 100%     | ✓         | 0.2                       | Mom                     | 0.2                     | Mom                     | 0.5                     | Mom                     | 789                 | 6,514          | Gain | 5,955                        | 9%                   | 2.6%   | 8.1%   | 4.5%          | 1.4  | 54.1         | 12.7         | 1.1          | 8.8          | 2.3          | 0%                 |  |
| 1942              | KANESAWA CO LTD (1942)    | Construction               | 1→1→1→2             | 2.3                            | 83%      | ✗         | -0.2                      | Rev                     | -0.2                    | Rev                     | -0.5                    | Rev                     | 1,250               | 6,088          | Pain | 6,378                        | -3%                  | -16.2% | -0.9%  | -0.0%         | 0.8  | 30.8         | 17.7         | 3.1          | 16.8         | 2.1          | 0%                 |  |
| 2854              | KOKA HOLDINGS (2854)      | Electric Appliances        | 90→7→4→3            | 2.2                            | 100%     | ✓         | 0.1                       | Mom                     | 0.4                     | Mom                     | 1.2                     | Mom                     | 20,021              | 51,290         | Gain | 27,030                       | 90%                  | 1.6%   | 56.7%  | 149.5%        | 2.0  | 103.1        | 6.7          | 20.0         | 51.9         | N/A          | 82%                |  |
| 5706              | MIYUKI KINOKUNI (5706)    | Nonferrous Metals          | 28→13→6→4           | 2.2                            | 100%     | ✓         | 0.2                       | Mom                     | 0.6                     | Mom                     | 0.7                     | Mom                     | 2,547               | 44,360         | Gain | 33,854                       | 31%                  | -19.2% | 18.6%  | 41.8%         | 1.5  | 88.1         | 30.5         | 6.2          | 24.5         | 0.6          | 47%                |  |
| 7167              | MEBURI FINANCIAL (7167)   | Banks                      | 13→4→6→5            | 2.2                            | 100%     | ✓         | 0.2                       | Mom                     | 0.2                     | Mom                     | 0.4                     | Mom                     | 1,312               | 1,386          | Gain | 1,274                        | 9%                   | 1.4%   | 8.1%   | 5.0%          | 1.3  | 45.7         | 13.0         | 1.2          | 8.2          | 2.9          | -                  |  |
| 6383              | DAIFUKU CO LTD (6383)     | Machinery                  | 4→5→3→6             | 2.2                            | 99%      | ✓         | -0.1                      | Rev                     | 0.2                     | Mom                     | 0.1                     | Mom                     | 2,562               | 6,746          | Gain | 6,170                        | 9%                   | -12.9% | 5.3%   | 6.7%          | 1.3  | 50.5         | 28.0         | 5.4          | 19.0         | 1.2          | 73%                |  |
| 5333              | NGK CORP (5333)           | Glass & Ceramics Products  | 20→20→7→7           | 2.3                            | 100%     | ✓         | 0.1                       | Mom                     | 0.6                     | Mom                     | 0.5                     | Mom                     | 1,594               | 5,456          | Gain | 4,493                        | 21%                  | -4.8%  | 20.1%  | 26.3%         | 1.1  | 45.7         | 19.0         | 1.9          | 7.8          | 1.9          | 78%                |  |
| 8359              | HACHIJUNI NAGANO (8359)   | Banks                      | 26→6→10→8           | 2.3                            | 100%     | ✓         | 0.2                       | Mom                     | 0.2                     | Mom                     | 0.6                     | Mom                     | 1,136               | 2,301          | Gain | 2,085                        | 10%                  | -2.4%  | 13.0%  | 10.0%         | 1.1  | 54.8         | 14.5         | 0.9          | 6.1          | 2.8          | 0%                 |  |
| 8334              | QUINBA BANK LTD (8334)    | Banks                      | 19→5→9→9            | 2.3                            | 99%      | ✓         | 0.1                       | Mom                     | -0.1                    | Rev                     | 0.4                     | Mom                     | 914                 | 2,309          | Gain | 2,135                        | 8%                   | 5.4%   | 9.0%   | 4.7%          | 1.2  | 30.9         | 13.6         | 1.4          | 10.0         | 3.0          | 0%                 |  |
| 6903              | MITSUBI ELEC CORP (6903)  | Electric Appliances        | 22→10→13→10         | 2.1                            | 99%      | ✓         | 0.0                       | Rev                     | 0.2                     | Mom                     | 0.3                     | Mom                     | 12,145              | 5,747          | Pain | 5,800                        | -1%                  | -11.1% | -4.7%  | -1.8%         | 1.3  | 47.5         | 23.9         | 2.6          | 9.7          | 1.0          | 49%                |  |
| 5802              | SUMITOMO ELEC N (5802)    | Nonferrous Metals          | 13→4→11→11          | 2.1                            | 100%     | ✓         | 0.0                       | Mom                     | 0.2                     | Mom                     | 0.3                     | Mom                     | 8,356               | 10,525         | Gain | 8,659                        | 22%                  | -17.6% | 2.2%   | 8.0%          | 1.6  | 84.1         | 25.3         | 3.0          | 14.7         | 0.4          | 62%                |  |
| 5803              | FUKUKURA LTD (5803)       | Nonferrous Metals          | 16→15→6→12          | 2.1                            | 96%      | ✓         | -0.2                      | Rev                     | 0.3                     | Mom                     | 0.2                     | Mom                     | 7,824               | 4,295          | Pain | 5,310                        | -19%                 | -45.3% | -2.1%  | 12.5%         | 1.9  | 107.5        | 33.7         | 12.7         | 32.5         | 0.9          | 77%                |  |
| 4004              | RESONAC HOLDINGS (4004)   | Chemicals                  | 95→26→15→13         | 2.1                            | 100%     | ✓         | 0.1                       | Mom                     | 0.5                     | Mom                     | 0.9                     | Mom                     | 2,942               | 15,910         | Gain | 12,174                       | 31%                  | -10.7% | 19.4%  | 41.8%         | 1.5  | 78.3         | 29.7         | 4.1          | 5.3          | 0.4          | 57%                |  |
| 6856              | HORIBA LTD (6856)         | Electric Appliances        | 76→10→20→14         | 1.9                            | 100%     | ✓         | 0.2                       | Mom                     | 0.7                     | Mom                     | 0.8                     | Mom                     | 1,041               | 24,645         | Gain | 21,148                       | 17%                  | 1.4%   | 16.5%  | 28.0%         | 1.3  | 43.7         | 21.6         | 3.0          | 11.6         | 2.0          | 77%                |  |
| 5831              | SHIZUKA FINANCIAL (5831)  | Banks                      | 35→30→22→15         | 1.9                            | 100%     | ✓         | 0.2                       | Mom                     | 0.3                     | Mom                     | 0.4                     | Mom                     | 1,724               | 2,972          | Gain | 2,834                        | 5%                   | 2.7%   | 9.0%   | -4.7%         | 1.1  | 49.2         | 15.0         | 1.3          | 7.5          | 3.3          | 0%                 |  |
| 9072              | NIKKON HOLDINGS (9072)    | Land Transportation        | 10→11→18→16         | 1.9                            | 97%      | ✓         | 0.1                       | Mom                     | 0.0                     | Mom                     | -0.2                    | Rev                     | 730                 | 5,772          | Gain | 4,381                        | 32%                  | 8.3%   | 17.5%  | 48.6%         | 0.5  | 63.7         | 35.5         | 2.8          | 7.6          | 1.9          | 18%                |  |
| 4062              | BIDEN CO LTD (4062)       | Electric Appliances        | 62→46→21→17         | 1.9                            | 100%     | ✓         | 0.2                       | Mom                     | 0.4                     | Mom                     | 0.5                     | Mom                     | 5,406               | 15,940         | Gain | 10,710                       | 49%                  | -8.3%  | 48.1%  | 70.8%         | 1.5  | 92.7         | 66.3         | 7.9          | 12.2         | 0.2          | 66%                |  |
| 1417              | MIRAIT ONE CORP (1417)    | Construction               | 11→27→17→18         | 1.9                            | 96%      | ✓         | 0.0                       | Rev                     | 0.3                     | Mom                     | -0.2                    | Rev                     | 351                 | 3,946          | Pain | 3,697                        | -1%                  | -7.3%  | -1.3%  | -1.1%         | 0.8  | 34.2         | 13.4         | 1.2          | 8.6          | 2.5          | -                  |  |
| 1803              | SHIMIZU CORP (1803)       | Construction               | 2→4→11→19           | 1.9                            | 88%      | ✗         | -0.3                      | Rev                     | -0.1                    | Rev                     | -0.6                    | Rev                     | 1,860               | 2,596          | Pain | 3,046                        | -15%                 | -17.0% | -6.4%  | -24.8%        | 0.9  | 80.9         | 12.8         | 1.8          | 13.8         | 3.0          | 0%                 |  |
| 5844              | KYOTO FINANCIAL (5844)    | Banks                      | 51→28→24→20         | 1.8                            | 94%      | ✓         | 0.1                       | Mom                     | 0.2                     | Mom                     | 0.3                     | Mom                     | 1,321               | 4,385          | Gain | 4,184                        | 5%                   | -2.2%  | 1.2%   | 18.6%         | 1.2  | 47.9         | 17.8         | 1.1          | 8.7          | 2.4          | 0%                 |  |
| 7186              | YOKOHAMA FINANCIAL (7186) | Banks                      | 77→53→28→21         | 1.7                            | 100%     | ✓         | 0.1                       | Mom                     | 0.4                     | Mom                     | 0.4                     | Mom                     | 1,892               | 1,653          | Gain | 1,535                        | 8%                   | 2.2%   | 12.7%  | -2.7%         | 1.3  | 47.8         | 14.6         | 1.3          | 7.9          | 2.8          | 0%                 |  |
| 1951              | EXEO GROUP INC (1951)     | Construction               | 24→24→16→22         | 1.7                            | 95%      | ✓         | -0.2                      | Rev                     | 0.1                     | Mom                     | 0.0                     | Rev                     | 573                 | 2,785          | Pain | 2,840                        | -2%                  | -6.4%  | -1.1%  | -3.2%         | 0.6  | 30.0         | 15.6         | 1.7          | 9.4          | 2.9          | 0%                 |  |
| 7337              | HIROGIN HOLDINGS (7337)   | Banks                      | 75→40→39→23         | 1.7                            | 94%      | ✓         | 0.2                       | Mom                     | 0.2                     | Mom                     | 0.4                     | Mom                     | 631                 | 2,068          | Gain | 1,879                        | 10%                  | 1.6%   | 12.4%  | 7.5%          | 1.2  | 43.0         | 12.2         | 1.1          | 8.1          | 3.4          | 0%                 |  |
| 6971              | KYOCERA CORP (6971)       | Electric Appliances        | 15→12→78→21         | 1.7                            | 100%     | ✓         | 0.2                       | Mom                     | 0.5                     | Mom                     | 0.7                     | Mom                     | 4,412               | 2,921          | Gain | 2,646                        | 19%                  | 1.4%   | 6.7%   | 10.0%         | 0.9  | 35.2         | 30.1         | 1.2          | 4.3          | 1.9          | 71%                |  |
| 8930              | SCORP HOLDINGS (8930)     | Insurance                  | 86→63→38→32         | 1.7                            | 91%      | ✓         | 0.2                       | Mom                     | 0.4                     | Mom                     | 0.4                     | Mom                     | 5,828               | 6,238          | Gain | 5,935                        | 5%                   | 2.5%   | 5.9%   | 5.7%          | 1.2  | 38.9         | 10.7         | 1.1          | 11.7         | 3.2          | 80%                |  |
| 8411              | MIZUHO FINANCIAL (8411)   | Banks                      | 43→22→26→26         | 1.7                            | 99%      | ✓         | 0.1                       | Mom                     | 0.0                     | Mom                     | 0.2                     | Mom                     | 17,056              | 6,982          | Gain | 6,825                        | 2%                   | -1.0%  | 5.0%   | -2.7%         | 1.4  | 52.5         | 12.5         | 1.5          | 11.5         | 2.1          | 55%                |  |
| 4186              | TOYOHI OKAWA KOGYO (4186) | Chemicals                  | 37→51→29→27         | 1.7                            | 100%     | ✓         | 0.1                       | Mom                     | 0.2                     | Mom                     | 0.1                     | Mom                     | 1,271               | 9,844          | Gain | 8,262                        | 20%                  | -11.7% | 6.4%   | 7.7%          | 1.5  | 72.6         | 28.3         | 5.1          | 17.3         | 0.8          | 85%                |  |
| 6754              | ANRITSU CORP (6754)       | Electric Appliances        | 47→39→27→28         | 1.7                            | 100%     | ✓         | 0.0                       | Mom                     | 0.2                     | Mom                     | 0.2                     | Mom                     | 595                 | 3,881          | Gain | 2,969                        | 48%                  | 4.6%   | 19.6%  | 67.8%         | 1.0  | 63.0         | 38.6         | 4.2          | 9.1          | 1.1          | 58%                |  |
| 8308              | RESONA HOLDINGS (8308)    | Banks                      | 72→19→31→29         | 1.7                            | 99%      | ✓         | 0.1                       | Mom                     | 0.0                     | Rev                     | 0.3                     | Mom                     | 4,680               | 2,029          | Gain | 1,870                        | 9%                   | -1.8%  | 6.0%   | 2.4%          | 1.3  | 51.1         | 14.5         | 1.6          | 9.2          | 1.8          | 0%                 |  |
| 5334              | NIITERRA CO LTD (5334)    | Glass & Ceramics Products  | 99→35→30→30         | 1.6                            | 100%     | ✓         | 0.0                       | Mom                     | 0.1                     | Mom                     | 0.5                     | Mom                     | 1,862               | 936            | Gain | 7,885                        | 19%                  | -3.6%  | 7.0%   | 19.9%         | 1.1  | 46.4         | 16.4         | 2.4          | 15.7         | 2.2          | 81%                |  |
| 5016              | JX ADVANCED META (5016)   | Nonferrous Metals          | 53→31→14→21         | 1.6                            | 93%      | ✓         | -0.3                      | Rev                     | 0.1                     | Mom                     | 0.2                     | Mom                     | 3,509               | 3,779          | Pain | 4,020                        | -6%                  | -18.7% | -21.8% | 11.5%         | 1.7  | 94.6         | 22.3         | 4.8          | 15.6         | 0.5          | 40%                |  |
| 8306              | MIYUBISHI UFJ F (8306)    | Banks                      | 65→34→42→32         | 1.6                            | 94%      | ✓         | 0.1                       | Mom                     | 0.1                     | Mom                     | 0.1                     | Mom                     | 35,959              | 3,030          | Gain | 2,858                        | 6%                   | 3.5%   | 7.0%   | 3.0%          | 1.3  | 38.5         | 13.4         | 1.5          | 11.3         | 3.2          | -                  |  |
| 9147              | NIPPON EXPRESS (9147)     | Land Transportation        | 190→50→19→13        | 1.6                            | 99%      | ✓         | -0.2                      | Rev                     | 0.2                     | Mom                     | 0.8                     | Mom                     | 1,171               | 4,018          | Gain | 3,879                        | 34%                  | 10.4%  | 25.2%  | 31.3%         | 0.6  | 31.6         | 19.3         | 1.4          | 0.7          | 2.1          | 40%                |  |
| 1721              | COMSYS HOLDINGS (1721)    | Construction               | 61→60→32→34         | 1.6                            | 100%     | ✓         | 0.1                       | Mom                     | 0.4                     | Mom                     | 0.2                     | Mom                     | 634                 | 5,370          | Pain | 5,455                        | -2%                  | -5.4%  | -3.4%  | -2.9%         | 0.7  | 37.9         | 16.1         | 1.5          | 9.4          | 2.5          | 0%                 |  |
| 6752              | PANASONIC HOLDN (6752)    | Electric Appliances        | 67→42→23→35         | 1.6                            | 99%      | ✓         | -0.1                      | Rev                     | 0.1                     | Mom                     | 0.3                     | Mom                     | 7,872               | 3,207          | Gain | 2,810                        | 14%                  | -2.1%  | 6.2%   | 27.7%         | 1.0  | 44.7         | 16.3         | 1.4          | 9.8          | 1.7          | 58%                |  |
| 8015              | TOYOTA TSUSHO (8015)      | Wholesale Trade            | 14→26→34→36         | 1.6                            | 90%      | ✓         | 0.0                       | Mom                     | 0.0                     | Rev                     | -0.4                    | Rev                     | 7,183               | 6,763          | Gain | 6,489                        | 4%                   | -8.6%  | 7.1%   | 0.4%          | 1.3  | 56.1         | 16.0         | 2.3          | 12.8         | 1.8          | 71%                |  |
| 8316              | SMFG (8316)               | Banks                      | 89→88→51→37         | 1.6                            | 95%      | ✓         | 0.2                       | Mom                     | 0.4                     | Mom                     | 0.3                     | Mom                     | 22,513              | 5,882          | Gain | 5,663                        | 4%                   | 0.5%   | 7.5%   | -1.4%         | 1.2  | 42.7         | 12.4         | 1.4          | 10.4         | 2.3          | -                  |  |
| 8060              | CANON MARKETING (8060)    | Wholesale Trade            | 49→45→43→38         | 1.5                            | 92%      | ✓         | 0.1                       | Mom                     | 0.1                     | Mom                     | 0.1                     | Mom                     | 834                 | 3,754          | Gain | 3,579                        | 5%                   | -0.5%  | 4.0%   | 9.6%          | 0.7  | 25.8         | 17.2         | 2.0          | 11.7         | 2.3          | 0%                 |  |
| 6901              | MURATA MFG CO (6901)      | Electric Appliances        | 298→100→71→38       | 1.5                            | 100%     | ✓         | 0.3                       | Mom                     | 0.1                     | Mom                     | 1.1                     | Mom                     | 12,100              | 6,194          | Gain | 4,454                        | 38%                  | 1.2%   | 28.1%  | 67.7%         | 1.3  | 94.1         | 35.0         | 4.1          | 8.8          | 1.1          | 92%                |  |
| 8309              | SUMITOMO MITSUI (8309)    | Banks                      | 92→72→59→40         | 1.5                            | 99%      | ✓         | 0.2                       | Mom                     | 0.3                     | Mom                     | 0.3                     | Mom                     | 3,943               | 5,643          | Gain | 5,410                        | 4%                   | -0.2%  | 6.8%   | 2.9%          | 1.2  | 38.8         | 11.1         | 1.1          | 9.5          | 0.8          | 34%                |  |
| 4026              | TAYO HOLDINGS (4026)      | Chemicals                  | 3→15→23→41          | 1.5                            | 92%      | ✗         | -0.1                      | Rev                     | -0.2                    | Rev                     | -0.9                    | Rev                     | 963                 | 4,915          | Pain | 5,252                        | -4%                  | 0.9%   | 0.2%   | -19.6%        | 0.6  | 37.3         | 22.1         | 4.6          | 22.0         | 1.7          | 65%                |  |
| 4272              | NIPPON KAYAKU (4272)      | Chemicals                  | 56→51→56→42         | 1.5                            | 100%     | ✓         | 0.1                       | Mom                     | 0.6                     | Mom                     | 0.1                     | Mom                     | 325                 | 2,030          | Gain | 1,842                        | 10%                  | -6.6%  | 14.0%  | 5.4%          | 0.8  | 50.2         | 13.4         | 1.1          | 9.0          | 3.3          | 55%                |  |
| 8341              | 77 BANK LTD (8341)        | Banks                      | 18→12→44→43         | 1.5                            | 64%      | ✓         | 0.0                       | Mom                     | -0.4                    | Rev                     | -0.4                    | Rev                     | 742                 | 3,226          | Gain | 3,066                        | 5%                   | 1.6%   | 5.1%   | -1.5%         | 1.2  | 48.7         | 11.9         | 1.0          | 8.5          | 3.2          | -                  |  |
| 5105              | NISHINIPPON CO (5105)     | Electric Appliances        | 104→50→48→44        | 1.5                            | 99%      | ✓         | 0.0                       | Mom                     | 0.0                     | Mom                     | 0.3                     | Mom                     | 388                 | 2,293          | Gain | 1,685                        | 36%                  | -5.2%  | 16.0%  | 33.9%         | 1.0  | 56.8         | 15.3         | 1.2          | 4.0          | 1.6          | 37%                |  |
| 6134              | FUJI CORP (6134)          | Machinery                  | 141→128→48→45       | 1.4                            | 100%     | ✓         | 0.1                       | Mom                     | 0.5                     | Mom                     | 0.4                     | Mom                     | 706                 | 7,220          | Gain | 5,359                        | 35%                  | 3.3%   | 21.0%  | 28.9%         | 1.1  | 60.3         | 21.8         | 2.7          | 7.0          | 2.6          | 89%                |  |
| 8031              | MIYUBI & CO (8031)        | Wholesale Trade            | 63→54→61→46         | 1.4                            | 94%      | ✓         | 0.2                       | Mom                     | 0.0                     | Mom                     | 0.1                     | Mom                     | 16,627              | 5,804          | Pain | 5,869                        | -1%                  | -5.3%  | 3.1%   | 5.4%          | 1.2  | 48.4         | 16.9         | 1.9          | 10.2         | 2.4          | 52%                |  |
| 7180              | KYUSHU FINANCIAL (7180)   | Banks                      | 148→87→70→47        | 1.4                            | 96%      | ✓         | 0.2                       | Mom                     | 0.3                     | Mom                     | 0.4                     | Mom                     | 658                 | 1,420          | Gain | 1,291                        | 11%                  | 6.5%   | 11.8%  | 7.7%          |      |              |              |              |              |              |                    |  |
| 7189              | NISHINIPPON FIN (7189)    | Banks                      | 63→56→41→48         | 1.4                            | 93%      | ✓         | 0.1                       | Mom                     | 0.0                     | Mom                     | 0.0                     | Mom</                   |                     |                |      |                              |                      |        |        |               |      |              |              |              |              |              |                    |  |

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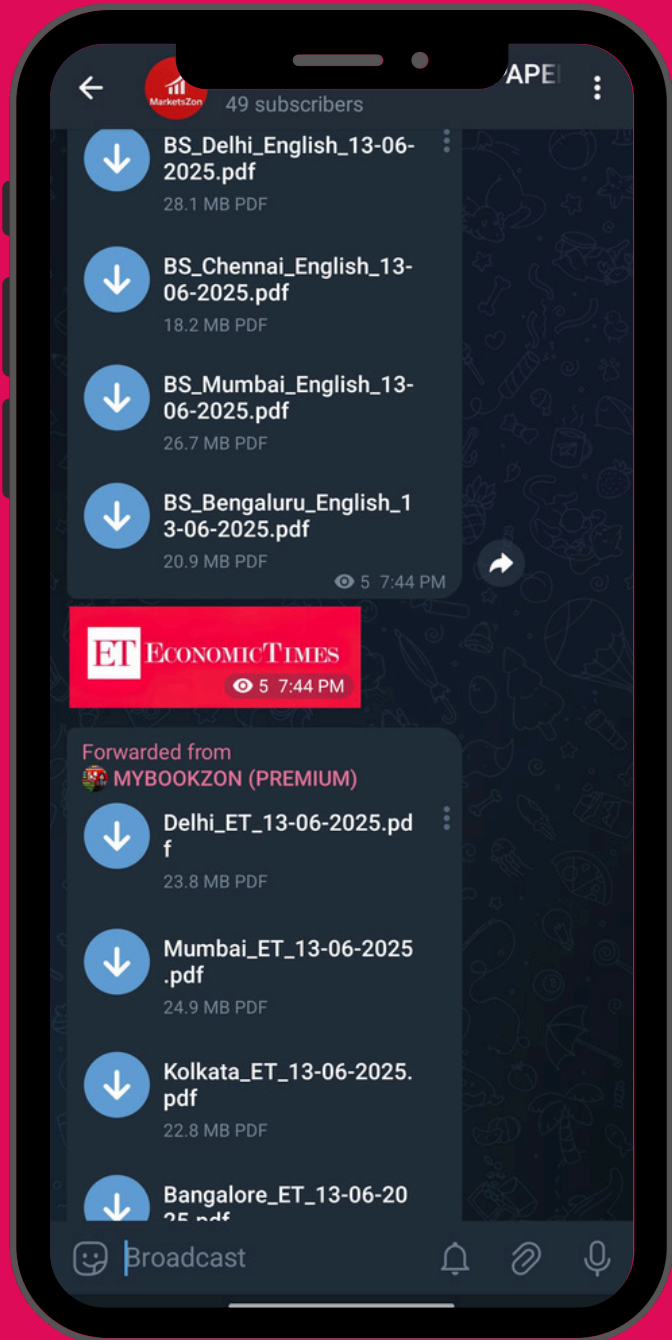
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